

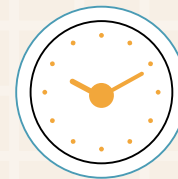


Pennywise



Wise Choices for Smart Spending

developed by **InfiniteLoop Team 11**



Deployment URL : <https://pennywise-frontend-rnxv.onrender.com>

GitHub : <https://github.com/anushkarajora11/Pennywise>

TA : Ujjwal Kajal



INTRODUCTION

- In today's digital era, managing **personal finances** has become an essential yet challenging task for individuals.
- Many people struggle to keep track of their daily expenses and savings, which often leads to poor financial planning.
- To address this problem, our project **PennyWise** is developed as an **Expense Tracking System** that provides a systematic and reliable way to manage financial activities.



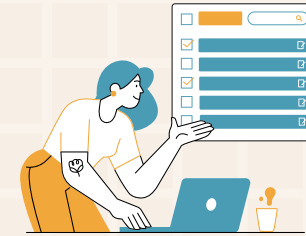
INTRODUCTION | System overview



- **PennyWise** enables users to record and categorize their income and expenses, helping them gain clear insights into their spending behavior. The system supports organized financial tracking, budget awareness, and better decision-making.
- By focusing on **simplicity** and **usability**, the application ensures that users can efficiently monitor and control their finances in their day-to-day life.



Need for the Software



- **Eliminate Manual Inefficiencies:** Existing tracking systems rely on manual calculations, making the generation of totals and proper business reports slow and resource-intensive.
- **Reduce Paperwork:** Documenting the flow of inward and outward supplies currently depends on physical paperwork, which is difficult to scale and organize.
- **Minimize Human Error:** Manually logging daily transactions, calculating GST, and updating product details is tedious and prone to costly inaccuracies.
- **Enable Active Budget Management:** Existing manual methods lack real-time spending alerts, making it incredibly difficult for users to track their expenses against strict budget limits and preventing proactive financial decisions.
- **Enhance Data Security:** A centralized digital database is required to securely store and easily retrieve sensitive financial information, replacing outdated physical books.

POTENTIAL USERS



College Students

Young Professionals

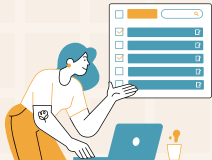


Salaried Employees

Families



Personal Budget Planners



Savings-Oriented Users



Major Requirements



Functional requirements:

User Authentication

- Sign up, login, OTP verification
- Forgot password with OTP

Transaction Management

- Add, edit, delete transactions
- Income & expense tracking
- Categorization (food, travel, etc.)

Analytics

- Graphs (spending trends, category-wise)
- Monthly/yearly insights
- CSV export

Alerts & Notifications

- Alerts when nearing/exceeding budget
- Notification panel updates

Budget Management

- Set category-wise budgets
- Track spending vs budget
- Real-time updates

Major Requirements



Non-Functional Requirements

Performance

- Fast loading dashboard
- Real-time updates

Security

- Encrypted login system
- OTP verification
- Protected user data

Usability

- Simple UI (sidebar navigation, dashboard)
- Easy to use for students & beginners

Reliability

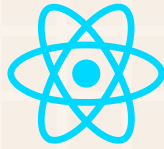
- Works consistently without crashes
- Accurate financial calculations

Scalability

- Can support more users and transactions

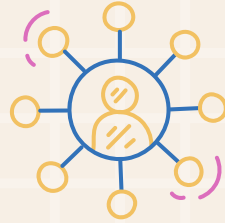


DESIGN

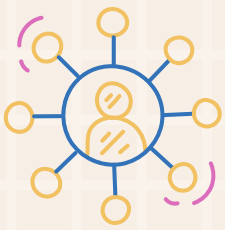


- MVC-inspired layered architecture.
- React-based view layer for page-wise screens like Records, Transactions, Category, Budget, Analytics, Alerts, and Profile.
- Express routes and controllers handle request flow, validation, and business logic.
- MySQL relational database used as the data layer for users, transactions, categories, budgets, alerts, and OTP records.
- Modular feature-based design with separate backend modules for alerts, analytics, budget, category, profile, transactions, and user.
- Security-focused design using JWT authentication middleware, OTP verification, and password hashing.

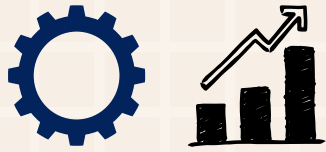
IMPLEMENTATION



- Frontend: React, HTML, CSS, JavaScript, and React Router.
- Backend: Node.js with Express.js for REST API development.
- Database: MySQL connected using mysql2.
- Analytics & visualization: Recharts for graphs, line charts, and pie charts.
- Authentication: bcrypt for password hashing, json web tokens for login sessions, OTP via email.
- Notifications/email: nodemailer with Gmail SMTP or preview mail flow for OTPs and budget alerts.
- Testing and collaboration: Jest, Supertest, Git, and GitHub.

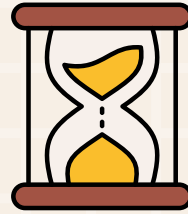


FUTURE IMPLEMENTATION



- AI-based Expense Prediction that uses past spending data to predict future expenses and suggest budgets to prevent overspending.
- Smart Savings Recommendations based on user spending patterns
- Bank API Integration that automatically fetches transactions from bank accounts to reduce manual data entry.
- Multi-user Collaboration for enabling family or group expense tracking with a shared budgeting system.
- Bill & EMI Reminders that notifies users before due dates with timely alerts to avoid missed payments.

LEARNINGS



Technical Learnings

Developing full-stack web applications using both frontend and backend technologies

Integrating APIs to connect frontend applications with backend services

Managing and optimizing databases for efficient data storage and retrieval

Using version control systems like Git and GitHub for collaborative development

Building frontend applications using frameworks like React and developing backend services using Node.js

Project Learnings

Coordinated effectively with team members to ensure smooth collaboration and task completion.

Managed time and priorities efficiently to complete the project before deadlines.

Prioritizing tasks based on project requirements and deadlines

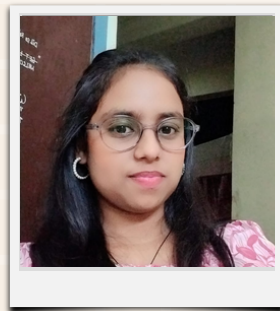
Identifying and solving problems encountered during project development

Communicating clearly and effectively within the team

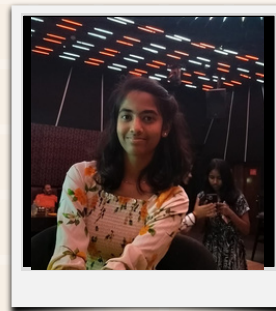
Our Team



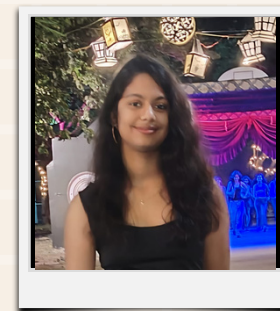
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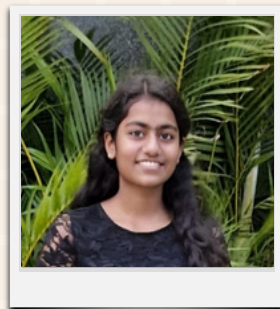
Anushika
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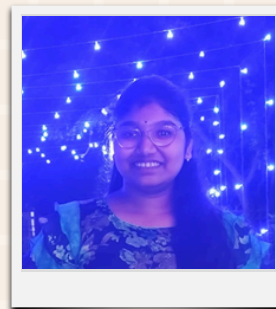
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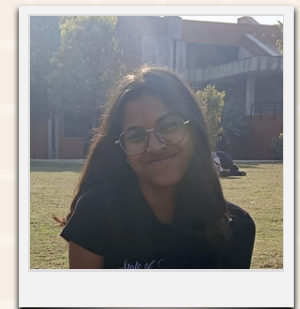
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**THANK
YOU**

